

Which limits of Newton's construction reach the zero branch

The thesis under test (user, 2026-10-01): Newton's mechanics exists only with a positive action scale, no zero branch is continuously reachable, and the family of theories has a gap in the sense of the mass gap. This note states the thesis as a claim about limits and proves, in the simplest cases, which of Newton's own limits reach the zero theory and which do not. Result: with Gaussian preparations and Gaussian marks, Galileo's comparison has a record family that is the same for every value of the constant (Theorem 1), so there is no gap there. For a determinate preparation, one definite velocity at each place, the complete coordinate record has no limit as the constant tends to zero once the preparation has folded, or a limit different from Newton's, while every smeared record converges to Newton's branch-weighted density (Theorem 2, any finite partition; Theorem 3, exactly, for inertial motion with a graded velocity after its fold time). The gap thesis therefore holds in the topology of complete records of determinate preparations, beyond their first fold, and fails in the topology of smeared records or for statistical preparations. Refinement with records is the one Newtonian limit in which the zero theory is the regular side. Proofs are written derivations by Claude Fable (2026-10-01), unrefereed; the stationary phase and almost-periodicity facts used are standard and named.

Throughout, h denotes the reduced phase constant, as in the 1998-conjecture note; the recording notes write $\kappa = h/2$.

1. The thesis as a statement about limits

For each $h > 0$ let T_h be the theory with phase constant h : its dynamics and its record law. The dilation $h \mapsto \mu h$ is an isomorphism (fifth-postulate note, Theorem B and §7; Rivero 1998, eq. (10), full read), so $\{T_h : h > 0\}$ is one theory in different units. The zero theory T_0 is Newton's. The thesis says that T_0 is not the limit of T_h as $h \rightarrow 0$ in any topology intrinsic to Newton's construction: the scale is generated by the construction, and zero is isolated. This is the mechanical form of a mass gap, where the continuum limit generates the scale and the massless theory is not reached by the coupling.

Known results fix what the thesis can mean. The limit $h \rightarrow 0$ is singular, with phases $e^{iS/h}$ that have no pointwise limit [Berry2002; metadata]. Yet in weak topologies and for finite times the classical theory is reached: coherent states follow classical trajectories for finite times [Hepp1974; metadata], Egorov's theorem transports smooth observables classically up to $O(h)$ for finite times, and Wigner measures give the weak limits of densities [LionsPaul1993; metadata]. So the thesis is false for smeared records at finite time, and can be true only in stronger topologies. Newton's construction supplies three candidates: complete records, the densities themselves rather than their smearings; the all-time limit; and refinement, with or without records at the inserted times. The rest of the note tests each.

Deformation families cannot see a gap. The Moyal family, the Fisher field law and its sheet realizations (score-constrained note, §§6–8) are continuous in the constant by construction and contain the zero branch. The tangent groupoid glues the $\varepsilon = 0$ fibre continuously to the $\varepsilon > 0$ fibres (fifth-postulate note, §7), so the kinematics is continuous at zero as well. A gap, if it exists, lives in the dynamics and the records.

2. Theorem 1: Galileo's comparison has no gap

Theorem 1 (Gaussian preparations, quadratic forces, Gaussian marks). Let $H = p^2/2m + V$ with $\deg V \leq 2$ (inertia, uniform gravity, Hooke's force). For each admissible $h \geq 0$ prepare the body with the Gaussian phase-space law $N(z_0, \Sigma)$, $\Sigma > 0$ fixed; for $h > 0$ this is the Gaussian state with that Wigner function, which exists iff $\det \Sigma \geq h^2/4$. Mark the body at times t_j by Gaussian coordinate copies of widths σ_j , each read or forgotten. Then the joint law of all read records and of the final coordinate is, for every admissible h , the classical Gaussian process

of T_0 with one addition: each mark, read or not, adds to the body momentum an independent centred Gaussian kick of variance $h^2/(4\sigma_j^2)$. The family is continuous in h and equals T_0 at $h = 0$.

Proof. For $\deg V \leq 2$ every Moyal flow is the classical affine symplectic flow, and the Wigner function is transported classically (fifth-postulate note, Theorem A(b)). A coordinate copy is generated by $q\pi$, quadratic in the joint canonical variables, so the same holds for the body–pointer shear. A Gaussian Wigner function of a product state is a positive density, and reading the pointer coordinate is classical conditioning of that density. The pointer’s momentum π has variance $h^2/(4\sigma_j^2)$ and is independent of its coordinate for a minimal Gaussian pointer, so the shear transfers to the body a kick of that variance whether or not the coordinate is read; this is Proposition 1 of the parallel-move note for the forgotten case, and the read case adds the classical Gaussian conditioning on $r = q + z$. Composition in time is the classical flow between marks. The variance $h^2/(4\sigma_j^2)$ is continuous in h and zero at $h = 0$. $\square$

So on the comparison as the Planck paper sets it, the record family is constant in h up to continuous kicks. The zero branch is reached in every topology: there is no gap in Galileo’s parabola with statistical preparations. The same holds for refinement with records at every inserted time: the fixed- h limit exists iff $\sum_j \sigma_j^{-2} < \infty$ (parallel-move note), the $h = 0$ limit always exists, and the $h \rightarrow 0$ limit at any fixed partition is continuous. In this limit the zero theory is the regular side and $h > 0$ the singular one.

3. Theorem 2: determinate preparations with several paths

Take the symmetric discrete action of the 1998-conjecture note, (1) there, on a partition with N cells of lengths τ_j , smooth V , fixed initial point $q_0 = x$, record variable $y = q_N$, and internal vertices $z = (q_1, \dots, q_{N-1}) \in \mathbb{R}^d$, $d = N - 1$. With a cutoff $\chi \in C_c^\infty(\mathbb{R}^d)$ and a smooth amplitude O , the halved functional with the record variable kept free is

$$A_h(y) = (2\pi h)^{-d/2} \int \chi(z) O(z) e^{iS_\pi(x, z, y)/h} dz. \quad (1)$$

Its modulus squared $|A_h(y)|^2$ is the complete coordinate record at the final time; $\int \varphi(y) |A_h(y)|^2 dy$ with $\varphi \in C_c^\infty$ is a smeared record.

Hypothesis (several nondegenerate paths). On an open interval Y of record values, $S_\pi(x, \cdot, y)$ has exactly $K \geq 2$ critical points $z_a(y)$ in $\text{supp } \chi$, all nondegenerate, with $\chi = 1$ near each. By the implicit function theorem they depend smoothly on y . Write

$$S_a(y) = S_\pi(x, z_a(y), y), \quad c_a(y) = \frac{O(z_a(y)) e^{i\pi\sigma_a/4}}{|\det S''_\pi(z_a(y))|^{1/2}}, \quad (2)$$

with σ_a the Hessian signature.

Lemma (final momenta are distinct). $p_a(y) := S'_a(y)$ equals $m(y - q_{N-1}^{(a)}(y))/\tau_{N-1} - \frac{1}{2}\tau_{N-1}V'(y)$, and $p_a(y) \neq p_b(y)$ for $a \neq b$.

Proof. The envelope formula gives $S'_a(y) = \partial_y S_\pi$ at the critical point, which is the displayed expression. The discrete Euler–Lagrange equation at vertex j , $m(q_{j+1} - q_j)/\tau_j - m(q_j - q_{j-1})/\tau_{j-1} + \frac{1}{2}(\tau_{j-1} + \tau_j)V'(q_j) = 0$, is linear in q_{j-1} with coefficient $m/\tau_{j-1} \neq 0$. So a critical point is determined by (q_{N-1}, q_N) by backward recursion, two distinct critical points have distinct q_{N-1} , and the momenta differ. $\square$

Theorem 2. Under the hypothesis, uniformly for y in compact subsets of Y ,

$$|A_h(y)|^2 = \sum_a |c_a(y)|^2 + \sum_{a < b} 2|c_a(y)c_b(y)| \cos\left(\frac{S_a(y) - S_b(y)}{h} + \theta_{ab}(y)\right) + O(h). \quad (3)$$

(i) *Complete record.* Fix $y \in Y$. The limit of $|A_h(y)|^2$ as $h \downarrow 0$ exists only if, for every nonzero value ω of $S_a(y) - S_b(y)$, the amplitudes $|c_a c_b| e^{i\theta_{ab}}$ of the pairs with that action difference sum to

zero. For $K = 2$ with $S_1(y) \neq S_2(y)$ this means $c_1(y)c_2(y) = 0$. When the limit exists it equals $\sum_a |c_a|^2$ plus the cross terms of pairs with equal actions, which is Newton's branch-weighted density $D_{0,\pi}(|O|^2) = \sum_a |O(z_a)|^2 / |\det S''_\pi(z_a)|$ only if those cross terms vanish too. (ii) *Smeared record*. For every $\varphi \in C_c^\infty(Y)$,

$$\int \varphi(y) |A_h(y)|^2 dy \longrightarrow \sum_a \int \varphi(y) \frac{|O(z_a(y))|^2}{|\det S''_\pi(z_a(y))|} dy \quad (h \downarrow 0). \quad (4)$$

Proof. Equation (3) is the stationary phase expansion of (1) with nondegenerate critical points, (3) of the 1998-conjecture note, whose remainder is uniform in the parameter y on compact sets where the critical points stay nondegenerate (Hörmander's stationary phase theorem with parameters; standard). Expanding the modulus squared gives (3) with $\theta_{ab} = \pi(\sigma_a - \sigma_b)/4 + \arg O_a - \arg O_b$.

- (i) Put $s = 1/h$. The cross sum in (3) is an almost periodic function $f(s)$ in Bohr's sense, a finite trigonometric sum. If $|A_h(y)|^2$ converges as $h \downarrow 0$ then $f(s)$ converges as $s \rightarrow \infty$. An almost periodic function with a limit at infinity is constant: for any s_0 and $\delta > 0$ the set of δ -almost periods is relatively dense, so there are $T_n \rightarrow \infty$ with $|f(s_0 + T_n) - f(s_0)| < \delta$, and the limit forces $f(s_0) = \lim f$. A constant trigonometric sum has zero coefficient at every nonzero frequency: the Bohr mean of $f(s)e^{-i\omega s}$ is that coefficient. This proves the condition, and the value of the limit when it exists.
- (ii) In the cross term with indices $a \neq b$ the phase $\psi_{ab} = S_a - S_b$ has $\psi'_{ab} = p_a - p_b \neq 0$ on Y by the Lemma. One integration by parts, $\int g e^{i\psi/h} dy = ih \int (g/\psi')' e^{i\psi/h} dy$ for $g \in C_c^\infty(Y)$, bounds the term by $O(h)$. The diagonal terms and the uniform remainder give (4). $\square$

Corollary (finite resolution). Let $\varphi \geq 0$ be smooth with compact support and $\int \varphi = 1$, and let $\varphi_\delta(y) = \delta^{-1} \varphi((y - y_0)/\delta)$ be a record of resolution δ at $y_0 \in Y$. With $\Delta p > 0$ the minimum of $|p_a - p_b|$ over pairs $a \neq b$ on the support,

$$\int \varphi_\delta |A_h|^2 dy = \sum_a \int \varphi_\delta \frac{|O(z_a)|^2}{|\det S''_\pi(z_a)|} dy + O\left(\frac{h}{\delta \Delta p}\right) + O(h). \quad (4')$$

A record of resolution δ reaches Newton's density iff $h/(\delta \Delta p) \rightarrow 0$: the fringes, of spacing $h/\Delta p$, must be unresolved. The complete-record topology of (i) is the order of limits in which $\delta \rightarrow 0$ before $h \rightarrow 0$; the smeared topology of (ii) is the opposite order.

Proof. In the integration by parts of (ii), the derivative of φ_δ carries a factor δ^{-1} relative to φ_δ itself, and $\int \varphi_\delta = 1$; the remaining terms are as before. $\square$

Thus the complete record of a determinate preparation with several paths does not reach Newton's density as $h \downarrow 0$, by non-convergence in the generic case of distinct actions and by a wrong limit in the symmetric case of equal actions, while its smearing always does. Theorem 1 of the 1998-conjecture note is the case $d = 1$, $K = 3$, $y = 0$ of (i), including the retained cross term of the two mirror paths; smearing in y removes that term by (ii), because the mirror paths arrive at $y = 0$ with opposite momenta.

3b. The gap as a number: fringe visibility

A mass gap is a number that is positive for every value of the coupling and zero for the free theory. The dilation $h \mapsto \mu h$ forbids any scale from playing that role here, so the invariant must be dimensionless. It is the contrast of the fringes in (3).

Proposition (visibility invariant). Let $y_0 \in Y$ be a record point with two branches, $K = 2$, and write $\rho_a = |c_a(y_0)|^2 = |O(z_a)|^2 / |\det S''_\pi(z_a)|$ for Newton's branch densities there,

$\Delta p = |p_1 - p_2|(y_0)$. (i) For every $h > 0$ the complete record near y_0 oscillates with fringe spacing $2\pi h/\Delta p + O(h^2)$ and visibility

$$V = \frac{\max |A_h|^2 - \min |A_h|^2}{\max |A_h|^2 + \min |A_h|^2} = \frac{2\sqrt{\rho_1\rho_2}}{\rho_1 + \rho_2} + O(h), \quad (\text{V1})$$

the maximum and minimum taken over one fringe. The leading term is fixed by Newton's branch densities and is independent of h . At $h = 0$ the record is $\rho_1 + \rho_2$ and the visibility is zero. (ii) The limit set of $|A_h(y_0)|^2$ as $h \downarrow 0$ is the interval $(\rho_1 + \rho_2)[1 - V, 1 + V]$. (iii) For a record of resolution δ with kernel φ , and δ small enough that the fringe phase is affine across the window $(\delta^2|p'_1 - p'_2| \ll h)$, the visibility is

$$V_\delta = V |\hat{\varphi}(\delta\Delta p/h)| + O(h), \quad \hat{\varphi}(\xi) = \int \varphi(u) e^{i\xi u} du, \quad (\text{V2})$$

so a Gaussian kernel gives $V_\delta = V \exp[-(\delta\Delta p/h)^2/2]$. The contrast is seen iff $h \gg \delta\Delta p$, and the two orders of limits of the Corollary give V and 0.

Proof. (i) By (3) with $K = 2$, $|A_h(y)|^2 = \rho_1 + \rho_2 + 2\sqrt{\rho_1\rho_2} \cos(\psi(y)/h + \theta) + O(h)$ with $\psi = S_1 - S_2$ and $\psi' = p_1 - p_2 \neq 0$, uniformly near y_0 . Over one fringe the cosine runs through ± 1 , which gives the spacing, the extremes $\rho_1 + \rho_2 \pm 2\sqrt{\rho_1\rho_2}$ and (V1). (ii) $\psi(y_0) \neq 0$ generically, and $\psi(y_0)/h$ is continuous and unbounded in h , so its residue mod 2π takes every value along sequences $h \downarrow 0$; the cosine takes every value in $[-1, 1]$. If $\psi(y_0) = 0$ the cross term is constant and the limit is the top of the interval. (iii) Integrate the cross term against φ_δ : with $\psi(y) = \psi(y_0) + \Delta p(y - y_0) + O(\delta^2|\psi''|)$ on the window, the integral is $2\sqrt{\rho_1\rho_2} \operatorname{Re}[e^{i\psi(y_0)/h + i\theta} \hat{\varphi}(\delta\Delta p/h)]$ up to the stated error; the diagonal terms are unchanged. $\square$

With $K \geq 3$ branches, as after a fold, each pair carries its own contrast $2\sqrt{\rho_a\rho_b}/\sum_c \rho_c$ at its own spacing $2\pi h/|p_a - p_b|$. For inertia with graded velocity (Theorem 3) the branch densities are $a(q_a)^2/|1 + p'_0(q_a)t/m|$; for the Kepler swarm (Theorem 4) they are $a(\phi'_a)^2/|1 + I'_0\Omega't|_a$. In every case the invariant is computed from Newton's multi-stream density and is positive wherever two streams cross, for every $h > 0$, and zero at $h = 0$. That is the gap: a contrast rather than a scale, discontinuous at zero, and invisible at any fixed resolution once h falls below $\delta\Delta p$.

3c. Imprecision actions at both ends

The record is one end of the comparison; the preparation is the other. A determinate swarm attached to the $h > 0$ theory as one state (5) has no momentum thickness; a real swarm has one, and so does the same classical swarm attached as a mixture of localized states, whose thickness is of order $\sqrt{h}$. The following proposition says what a thickness does, in the exact free setting of §4, and the general case follows by the same envelope identity.

Proposition (preparation thickness). Shift the initial curve by a momentum η , $\psi_0^\eta = a e^{i(S_0 + \eta q)/h}$, and average the record over η with a density w of mean zero and width Δ . Let x be a record point with two arrivals $q_1 \neq q_2$, write $\Delta q_0 = |q_1 - q_2|$ for their launch separation, and let Δ be small enough that the branch densities and launch points are constant across the thickness up to $o(1)$ and $\Delta^2|\partial_\eta(q_1 - q_2)| \ll h$. Then the averaged record has visibility

$$V_\Delta = V |\hat{w}(\Delta q_0/h)| + O(h) + o(1), \quad \hat{w}(\xi) = \int w(\eta) e^{i\xi\eta} d\eta, \quad (\text{V3})$$

so a Gaussian thickness gives $V_\Delta = V \exp[-(\Delta\Delta q_0/h)^2/2]$.

Proof. The arrivals solve $q + (p_0(q) + \eta)t/m = x$ and the stationary actions are $\Phi_a^\eta = m(x - q_a)^2/2t + S_0(q_a) + \eta q_a$. By the envelope identity, $d\Phi_a^\eta/d\eta = q_a(\eta)$, so $\Phi_1^\eta - \Phi_2^\eta = \psi(0) + \eta(q_1 - q_2) + O(\eta^2|\partial_\eta(q_1 - q_2)|)$. Averaging the cross term of (3) against w under the stated hypotheses gives $2\sqrt{\rho_1\rho_2} \operatorname{Re}[e^{i\psi(0)/h + i\theta} \hat{w}((q_1 - q_2)/h)]$, and the diagonal terms change by $o(1)$. $\square$

With both imprecisions present and independent, the contrast is $V |\hat{\varphi}(\delta\Delta p/h)| |\hat{w}(\Delta\Delta q_0/h)|$. Each argument is an action divided by $\hbar$: the record imprecision $\delta\Delta p$, position resolution times the arrival momentum difference, and the preparation imprecision $\Delta\Delta q_0$, momentum thickness times the launch separation. The gap is seen iff both imprecision actions lie below $\hbar$. This is the floor the thesis asserts, and it runs opposite to the uncertainty relation: there $\hbar$ bounds imprecisions from below; here imprecisions must be pushed below $\hbar$ for the $\hbar > 0$ record to differ from Newton's. Newton's construction has exact preparations and exact records, both imprecision actions zero, so for every $\hbar > 0$ it sees the full contrast V , and at $\hbar = 0$ it sees none. The two attachments of the same classical swarm are now separated by a number: the mixture of localized states has $\Delta \sim \sqrt{\hbar}$, so $\Delta\Delta q_0/\hbar \rightarrow \infty$ and its contrast vanishes, while the state (5) has $\Delta = 0$. Determinacy at each $\hbar$, momentum thickness of order $\hbar$ at fixed position (the sheet separation $2\kappa \partial_q \log \rho$ of the score note) rather than $\sqrt{\hbar}$, is the premise that selects the first attachment.

4. Theorem 3: inertial motion with a graded velocity

Newton's simplest motion already shows the gap when the preparation is determinate and its velocity varies along it. Let $\hbar > 0$, $V = 0$, and prepare

$$\psi_0(q) = a(q)e^{iS_0(q)/\hbar}, \quad a \in C_c^\infty(\mathbb{R}) \text{ real}, \quad p_0 = S'_0. \quad (5)$$

Each particle at q has the definite velocity $p_0(q)/m$. The free propagator is exact, so

$$\psi_t(x) = \left(\frac{m}{2\pi i \hbar t}\right)^{1/2} \int a(q) e^{i\Phi_x(q)/\hbar} dq, \quad \Phi_x(q) = \frac{m(x-q)^2}{2t} + S_0(q). \quad (6)$$

Critical points of Φ_x are the arrivals $q + p_0(q)t/m = x$, and $\Phi'_x(q) = m/t + p'_0(q)$. Define the fold time

$$t_* = \frac{m}{\max(-p'_0)} \in (0, \infty], \quad (7)$$

with $t_* = \infty$ when $p'_0 \geq 0$ everywhere.

Theorem 3. (a) For $0 < t < t_*$ the arrival map $q \mapsto q + p_0(q)t/m$ is a diffeomorphism, every x has one arrival $q_*(x)$, and $|\psi_t(x)|^2 \rightarrow a(q_*(x))^2/(1 + p'_0(q_*(x))t/m)$ pointwise, the classical transported density: the complete record reaches the zero branch before the fold. (b) For $t > t_*$ there is an open interval of x with three nondegenerate arrivals $q_1 < q_2 < q_3$. Where a is nonzero at two or more of them, Theorem 2 applies with $d = 1$, $K = 3$: $|\psi_t(x)|^2$ has no limit as $\hbar \downarrow 0$ unless the cross amplitudes cancel, and every smearing in x converges to the classical three-branch density $\sum_a a(q_a)^2/|1 + p'_0(q_a)t/m|$. (c) At a fold point $x_c(t)$, where $\Phi'_x(q_c) = 0$ and $\Phi'''_x(q_c) \neq 0$, $|\psi_t(x_c)|^2$ grows like $\hbar^{-1/3}$: the record diverges with the classical caustic density.

Proof. (a) The derivative $1 + p'_0(q)t/m$ is positive for $t < t_*$, so the arrival map is strictly increasing and onto, and the single critical point is nondegenerate; one-dimensional stationary phase in (6) gives $|\psi_t|^2 \rightarrow a^2/|\Phi''_x| \cdot (m/t)$, which is the stated ratio. (b) For $t > t_*$ the derivative is negative on an open set, so the arrival map is not monotone; by the intermediate value theorem the preimage of each x in an open interval has three points, nondegenerate away from the two fold values. The amplitudes $c_a = a(q_a)e^{i\pi\sigma_a/4}/|\Phi''_x(q_a)|^{1/2}$ and the actions $\Phi_x(q_a)$ play the roles of (2); the final momenta $\partial_x \Phi_x(q_a) = m(x - q_a)/t = p_0(q_a)$ are distinct because the q_a are distinct and $x = q_a + p_0(q_a)t/m$. Theorem 2 applies. (c) At a degenerate critical point of order three the stationary phase integral scales as $\hbar^{1/3}$ against the prefactor $\hbar^{-1/2}$, giving $|\psi|^2 \sim \hbar^{-1/3}$ (Airy scaling; standard). $\square$

Newton's inertial motion, with a preparation in which faster particles start behind slower ones, folds at t_* . Before the fold the zero branch is reached by complete records; after it, never. The fringe spacing in x is $\hbar/|p_a - p_b|$, which tends to zero without the record ever settling. The classical multi-stream density is recovered only by smearing, an operation external to the determinate construction.

4b. Theorem 4: central forces on the radial cycle, Hooke against Kepler

Newton's central forces supply the case the thesis was made for. Fix the angular momentum $\ell > 0$. The radial motion is one degree of freedom with action $I = I_r$ and angle ϕ on the circle, Hamiltonian $E(I)$, frequency $\Omega(I) = E'(I)$ and radial period $T = 2\pi/\Omega$. The 1998-conjecture note, (10)–(11) gives, for Hooke's force (Book I, Proposition X), $E = \omega(2I + \ell)$, so $\Omega = 2\omega$ and $\Omega' = 0$, the isochrony; and for Kepler's,

$$E = -\frac{mk^2}{2(I + \ell)^2}, \quad \Omega = \frac{mk^2}{(I + \ell)^3}, \quad \Omega' = -\frac{3mk^2}{(I + \ell)^4}, \quad T = 2\pi\sqrt{\frac{ma^3}{k}}. \quad (8)$$

For Kepler the angle ϕ is the mean anomaly, which Kepler's equation turns into position on the ellipse: it is the record astronomy makes.

Model. Take $H_h = E(-ih\partial_\phi)$ on the circle with the twisted periodicity $\psi(\phi + 2\pi) = e^{2\pi i\alpha}\psi(\phi)$, α the Maslov shift of the radial cycle ($\alpha = \frac{1}{2}$), so that the spectrum is $E(h(n + \alpha))$, the Keller–Maslov values, which for Hooke and Kepler are the exact planar spectra ((9)–(11) of that note). Extend E smoothly outside the physical range of I ; for the preparations below this changes the evolved state by $O(h^\infty)$. The determinate preparation is a swarm,

$$\psi_0(\phi) = a(\phi)e^{iS_0(\phi)/h}, \quad a \in C_c^\infty((-\pi, \pi)), \quad I_0 = S'_0, \quad (9)$$

each body at ϕ on the orbit of action $I_0(\phi)$, with action spread $\Delta I = \max I_0 - \min I_0$ on the support and support length $|a| < 2\pi$. Poisson summation over the eigenvalue index gives exactly

$$\psi_t(\phi) = \sum_{k \in \mathbb{Z}} \frac{e^{-2\pi i k \alpha}}{2\pi h} \iint a(\phi') e^{i\Phi_k(\phi', I)/h} d\phi' dI, \quad \Phi_k = S_0(\phi') + I(\phi - \phi' + 2\pi k) - E(I)t. \quad (10)$$

The critical points of Φ_k are $I = I_0(\phi')$ and $\phi - \phi' + 2\pi k = \Omega(I)t$: the body prepared at ϕ' arrives at ϕ after k turns. The Hessian determinant is $-(1 + I'_0(\phi')\Omega'(I_0(\phi'))t)$. Define the fold time and the lapping time

$$t_* = \frac{1}{\max(-I'_0\Omega'(I_0))} \in (0, \infty], \quad t_{\text{lap}} = \frac{2\pi - |a|}{\max \Omega(I_0) - \min \Omega(I_0)} \in (0, \infty]. \quad (11)$$

Theorem 4. (a) *Hooke.* E is affine in I , the I -integral in (10) is a delta function, and for every $h > 0$ and every t

$$|\psi_t(\phi)|^2 = a(\phi - 2\omega t)^2 \pmod{2\pi} : \quad (12)$$

the swarm rotates rigidly, the complete anomaly record is the same for every h , and the zero branch is reached at all times. (b) *Kepler, or any $\Omega' \neq 0$.* For $t < \min(t_*, t_{\text{lap}})$ every ϕ has at most one arrival and $|\psi_t(\phi)|^2 \rightarrow a(\phi')^2/|1 + I'_0(\phi')\Omega'(I_0(\phi'))t|$ pointwise. For $t > t_{\text{lap}}$ some ϕ have two or more arrivals of different winding, and for $t > t_*$ three arrivals of the same winding; distinct arrivals at the same ϕ have distinct actions I_a , so Theorem 2 and its Corollary apply with ϕ as record variable and I as momentum: the complete anomaly record has no $h \rightarrow 0$ limit unless cross amplitudes cancel, every smeared record converges to the classical density $\sum_a a(\phi'_a)^2/|1 + I'_0\Omega't|_a$, and a record of angular resolution δ reaches it iff $h/(\delta \min |I_a - I_b|) \rightarrow 0$. (c) *Kepler times.* For a narrow swarm, $\Delta I \ll I + \ell$,

$$t_{\text{lap}} \simeq \frac{2\pi(I + \ell)^4}{3mk^2 \Delta I} = \frac{T(I + \ell)}{3 \Delta I} = \frac{T^2}{\Delta T}, \quad t_* = \frac{(I + \ell)^4}{3mk^2 \max I'_0} \quad (\max I'_0 > 0), \quad (13)$$

with $\Delta T = 3T\Delta I/(I + \ell)$ the spread of periods. Both are classical times, independent of h .

Proof. The eigenfunctions are $e^{i(n+\alpha)\phi}$ with eigenvalues $E(h(n + \alpha))$, and the preparation's coefficients are $c_n = (2\pi)^{-1} \int a e^{i[S_0(\phi') - h(n+\alpha)\phi']/h} d\phi'$. With $\nu = n + \alpha$ and $I = h\nu$ the evolved state is $\sum_n G(n + \alpha)$ for $G(\nu) = (2\pi)^{-1} \int a(\phi') e^{i[S_0(\phi') + h\nu(\phi - \phi') - E(h\nu)t]/h} d\phi'$, a Schwartz function

of ν : for $h\nu$ outside the range of I_0 the ϕ' -phase has no stationary point and integration by parts gives rapid decay, and derivatives in ν only add polynomial factors. Poisson summation, $\sum_n G(n + \alpha) = \sum_k e^{2\pi i k \alpha} \int G(\nu) e^{-2\pi i k \nu} d\nu$, with $k \rightarrow -k$ and $d\nu = dI/h$, is (10). (a) For $E = \omega(2I + \ell)$ the I -integral is $(2\pi h)^{-1} \int e^{iI(\phi - \phi' + 2\pi k - 2\omega t)/h} dI = \delta(\phi - \phi' + 2\pi k - 2\omega t)$, so $\psi_t(\phi) = \sum_k e^{-2\pi i k \alpha - i\omega \ell t/h} a(\phi + 2\pi k - 2\omega t) e^{iS_0(\phi + 2\pi k - 2\omega t)/h}$; one k contributes at each ϕ because $|a| < 2\pi$, and (12) follows. (b) The arrival map $\phi' \mapsto \phi' + \Omega(I_0(\phi'))t$ has derivative $1 + I'_0 \Omega' t > 0$ for $t < t_*$, so it is injective on the support, and its image has length $|a| + t(\max \Omega - \min \Omega) < 2\pi$ for $t < t_{\text{lap}}$, so no two windings overlap: one nondegenerate critical point at each ϕ of the image, none elsewhere up to $O(h^\infty)$. Two-dimensional stationary phase in (10) with the prefactor $(2\pi h)^{-1}$ gives the pointwise limit. For $t > t_{\text{lap}}$ the image length exceeds 2π and by continuity some ϕ is covered by two points of the support with different k ; for $t > t_*$ the map is not monotone and three arrivals of the same winding appear, as in Theorem 3. Two arrivals (ϕ'_1, k_1) , (ϕ'_2, k_2) at the same ϕ with the same action I would satisfy $\phi'_1 - \phi'_2 = 2\pi(k_1 - k_2)$, hence coincide because $|a| < 2\pi$; so distinct arrivals have distinct actions, and $\partial_\phi \Phi_k = I$ at the critical point is the record momentum of Theorem 2. (c) Insert (8) into (11): $|\Omega'| \Delta I$ is the frequency spread, $T = 2\pi(I + \ell)^3/(mk^2)$, and $T \propto (I + \ell)^3$ gives $\Delta T/T = 3\Delta I/(I + \ell)$. $\square$

Proposition X and Kepler's law thus fall on opposite sides of the thesis in the record Newton's astronomy actually makes. The isochronous force keeps the zero branch for all time; the Kepler force loses it for every swarm with a spread of periods, after the classical lapping time $T^2/\Delta T$, the time after which the fast bodies have gained a full turn on the slow ones. From then on the complete record of anomalies carries fringes of angular spacing $h/\Delta I$ that never settle. A stationary preparation, the whole ring at one action, is an eigenstate and reaches the zero branch trivially; the gap needs a localized swarm with a spread of actions. In the radial record r , by contrast, even Hooke's swarm has two branches with opposite radial momenta whenever it straddles a turning point, so there the gap appears for both forces during each turning passage; the anomaly record is the one that separates them.

5. All-time invariants

Two quantities already in the corpus show the other non-commuting pair of limits, $h \rightarrow 0$ against $t \rightarrow \infty$. For the free Gaussian, the expected number of sign reversals of the two-sheet realization over all time is $\pi\varphi(1)$ for every $h > 0$ and zero at $h = 0$ (recording note, (150)); the total phase $\arctan u$ of (143) there reaches $\pi/2$ for every $h > 0$ and is zero at $h = 0$. Both are limits taken at infinite time first; at fixed finite time both vanish continuously as $h \rightarrow 0$. The first is a hidden count and the second a phase, so neither is a record; their observable counterpart is the breakdown of the classical limit after the Ehrenfest time, which Theorem 3 exhibits at the fold time t_* for inertial motion. The all-time limit therefore supports the thesis only through folding, which is already Theorem 3.

6. What the thesis is, and what it needs

- **Galileo's comparison with statistical preparations has no gap.** Theorem 1 makes the record family constant in h . Any gap argument run on the parabola with Gaussian preparations and marks will fail, for the same reason the deformation families fail.
- **Determinate preparations that fold have the gap, in complete records.** Theorems 2 and 3 prove it on every finite partition and exactly for inertia with graded velocity; Theorem 4 proves it for Kepler's force in the anomaly record. Beyond Galileo the thesis holds in this topology.
- **Hooke and Kepler separate.** In the anomaly record, Proposition X's isochronous force keeps the zero branch for all time (Theorem 4(a)), while Kepler's force loses it for every swarm with a spread of periods after the classical lapping time $T^2/\Delta T$ (Theorem 4(b), (c)). The gap is a property of the force law.

- **Smeared records and statistical preparations reach zero.** This is the content of the weak-limit theorems named in §1, and of (4).
- **Refinement with records reverses the roles.** The zero theory is the regular side of that limit.
- **The thesis is an order of limits.** By the Corollary to Theorem 2, a record of resolution δ reaches Newton iff $h/(\delta\Delta p) \rightarrow 0$. “Complete records” means $\delta \rightarrow 0$ before $h \rightarrow 0$: the record limit is taken first, as the continuum limit is taken first in the mass gap.
- **The gap is a contrast, not a scale.** The fringe visibility $2\sqrt{\rho_1\rho_2}/(\rho_1 + \rho_2)$ of a complete record where two streams cross is fixed by Newton’s branch densities, independent of h for every $h > 0$, and zero at $h = 0$ (§3b). The dilation symmetry makes the invariant dimensionless where the mass gap is a scale.
- **Both ends must be exact.** The contrast is damped by $|\hat{\varphi}(\delta\Delta p/h)|$ at the record and by $|\hat{w}(\Delta\Delta q_0/h)|$ at the preparation (§3c): the gap is seen iff both imprecision actions lie below h . Newton’s construction has both zero; a classical swarm attached as a mixture of localized states has thickness $\sqrt{h}$ and shows no gap.

The thesis is therefore exactly as strong as two premises: that records are complete, the densities themselves rather than their smearings, and that preparations are determinate, one velocity at each place. Neither is foreign to Newton. A determinate preparation is the Newtonian default, every body with its own velocity; the statistical preparation of Theorem 1 is the extra assumption the Planck paper makes with its Gaussian marks. The second is the fifth postulate’s joint determinacy (fifth-postulate note) read as a preparation, Newton’s *velocitas ultima* assigned to every particle of the ensemble. The first is the records reading already required by Leibniz continuity. It is also Newton’s own topology. His limits are exact: the Section I scholium asks the reader to think of the vanishing quantities “semper diminuendas sine limite”, always diminishing without limit, and the geometry of the comparison has no floor at any resolution (Planck paper, §2, Proposition 1). Newton’s construction contains no apparatus resolution; the ultimate ratio is the exact record, and the resolution limit is taken inside the geometry before any constant enters. Read in that topology, the $h > 0$ family is Theorem 2(i)’s complete record. The smeared topology adds a resolution δ that the construction does not contain; the Planck paper’s marks are that modern addition. The gap claim and the Leibniz claim rest on the same reading of records, and the gap claim adds determinacy of the preparation. Under those two premises the statement “Newton exists only with a positive scale” has the precise form: the construction without branch data exists for every $h > 0$ and refines without extra data (1998-conjecture note, Theorem 2(b)), while at $h = 0$ it needs a branch rule (Theorem 1 there) and, after a fold, is not the limit of the $h > 0$ records (Theorems 2–3 here). With either premise dropped, the zero branch is reached.

7. Consequence for STATE

The gap thesis is located and has its first central-force case. It holds for complete records of determinate preparations beyond their first fold (Theorems 2–3), and in the anomaly record it separates the force laws: Hooke keeps the zero branch for all time, Kepler loses it after $T^2/\Delta T$ (Theorem 4). It fails for Galileo’s comparison with Gaussian preparations and marks (Theorem 1), for smeared records, and for refinement with records; a record of resolution δ reaches Newton iff $h/(\delta\Delta p) \rightarrow 0$. The gap has a number: the fringe visibility $2\sqrt{\rho_1\rho_2}/(\rho_1 + \rho_2)$ of a complete record where two streams cross, fixed by Newton’s branch densities, independent of h for every $h > 0$ and zero at $h = 0$ (§3b), seen iff the record and preparation imprecision actions $\delta\Delta p$ and $\Delta\Delta q_0$ both lie below h (§3c). Consequences: the Planck paper’s Gaussian parabola cannot carry the gap argument, which belongs to the Kepler swarm; the two premises, complete records as the record-limit-first order and determinate preparations as Newton’s default, are the gap’s hypotheses and the next things to defend, the first with the Leibniz records reading. Further

exact cases available: the radial record of a Hooke swarm at a turning passage, and the classical-first route read with ε in place of h , where Theorem 2(i) says the unselected construction at zero exists only for single-path problems. Reject a gap argument run inside a deformation family, on Gaussian preparations of the parabola, or on smeared records, and reject presenting the non-commutation of $h \rightarrow 0$ with $t \rightarrow \infty$ as a record without an observable.